

# Learning modern Java the playful way



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GOTO Copenhagen 2026

\$ who are we

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Senior Developer Advocate

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\$ who are you

**JAVA 24 DOESN'T  
PIN VIRTUAL THREADS**



**24? WE'RE  
STILL MIGRATING TO 21**



**21? WE'RE NOT ALLOWED  
TO USE RECORDS FROM 17**



**WAIT? YOU ARE  
ALLOWED TO GO BEYOND 8?**



# Agenda

- Various features from recent Java versions  
(not all of them, obviously)
- Demos  
(obviously)



## JDK 27

This release is the Reference Implementation of version 27 of the Java SE Platform, as specified by [JSR 402](#) in the Java Community Process.

The features and schedule of this release were proposed and tracked via the [JEP Process](#), as amended by the [JEP 2.0 proposal](#). The release was produced using the [JDK Release Process \(JEP 3\)](#).

### Status

JDK 27 reached [General Availability](#) on 15 September 2026. Production-ready binaries under the GPL are available from Oracle; binaries from other vendors will follow shortly.

### Features

- 523: Make G1 the Default Garbage Collector in All Environments
- 527: Post-Quantum Hybrid Key Exchange for TLS 1.3
- 531: Lazy Constants (Third Preview)
- 532: Primitive Types in Patterns, instanceof, and switch (Fifth Preview)
- 533: Structured Concurrency (Seventh Preview)
- 534: Compact Object Headers by Default
- 536: JFR In-Process Data Redaction
- 537: Vector API (Twelfth Incubator)
- 538: PEM Encodings of Cryptographic Objects (Third Preview)

### Schedule

- |            |                                                            |
|------------|------------------------------------------------------------|
| 2026/06/04 | <a href="#">Rampdown Phase One</a> (branch from main line) |
| 2026/07/16 | <a href="#">Rampdown Phase Two</a>                         |
| 2026/08/06 | <a href="#">Release Candidate Phase</a>                    |
| 2026/08/20 | <a href="#">Release Candidate Build</a>                    |
| 2026/09/15 | General Availability                                       |

- Compact Object Headers
- G1 garbage collector is default everywhere
- Preview features

## JDK 26

This release is the Reference Implementation of version 26 of the Java SE Platform, as specified by [JSR 401](#) in the Java Community Process.

The features and schedule of this release were proposed and tracked via the [JEP Process](#), as amended by the [JEP 2.0 proposal](#). The release was produced using the [JDK Release Process \(JEP 3\)](#).

### Status

JDK 26 reached [General Availability](#) on 17 March 2026. Production-ready binaries under the GPL are [available from Oracle](#); binaries from other vendors will follow shortly.

### Features

- 500: [Prepare to Make Final Mean Final](#)
- 504: [Remove the Applet API](#)
- 516: [Ahead-of-Time Object Caching with Any GC](#)
- 517: [HTTP/3 for the HTTP Client API](#)
- 522: [G1 GC: Improve Throughput by Reducing Synchronization](#)
- 524: [PEM Encodings of Cryptographic Objects \(Second Preview\)](#)
- 525: [Structured Concurrency \(Sixth Preview\)](#)
- 526: [Lazy Constants \(Second Preview\)](#)
- 529: [Vector API \(Eleventh Incubator\)](#)
- 530: [Primitive Types in Patterns, instanceof, and switch \(Fourth Preview\)](#)

### Schedule

- |            |                                                            |
|------------|------------------------------------------------------------|
| 2025/12/04 | <a href="#">Rampdown Phase One</a> (branch from main line) |
| 2026/01/15 | <a href="#">Rampdown Phase Two</a>                         |
| 2026/02/05 | <a href="#">Initial Release Candidate</a>                  |
| 2026/02/19 | <a href="#">Final Release Candidate</a>                    |
| 2026/03/17 | <a href="#">General Availability</a>                       |

- More pattern matching & data-oriented programming
- Concurrency stuff maturing
- Native interop API progress
- Stronger security defaults
- And more

## JDK 25

This release is the Reference Implementation of version 25 of the Java SE Platform, as specified by [JSR 400](#) in the Java Community Process.

JDK 25 reached [General Availability](#) on 16 September 2025. Production-ready binaries under the GPL are [available from Oracle](#); binaries from other vendors will follow shortly.

The features and schedule of this release were proposed and tracked via the [JEP Process](#), as amended by the [JEP 2.0 proposal](#). The release was produced using the [JDK Release Process \(JEP 3\)](#).

### Features

470: [PEM Encodings of Cryptographic Objects \(Preview\)](#)  
502: [Stable Values \(Preview\)](#)  
503: [Remove the 32-bit x86 Port](#)  
505: [Structured Concurrency \(Fifth Preview\)](#)  
506: [Scoped Values](#)  
507: [Primitive Types in Patterns, instanceof, and switch \(Third Preview\)](#)  
508: [Vector API \(Tenth Incubator\)](#)  
509: [JFR CPU-Time Profiling \(Experimental\)](#)  
510: [Key Derivation Function API](#)  
511: [Module Import Declarations](#)  
512: [Compact Source Files and Instance Main Methods](#)  
513: [Flexible Constructor Bodies](#)  
514: [Ahead-of-Time Command-Line Ergonomics](#)  
515: [Ahead-of-Time Method Profiling](#)  
518: [JFR Cooperative Sampling](#)  
519: [Compact Object Headers](#)  
520: [JFR Method Timing & Tracing](#)  
521: [Generational Shenandoah](#)

### Schedule

|            |                                                            |
|------------|------------------------------------------------------------|
| 2025/06/05 | <a href="#">Rampdown Phase One (branch from main line)</a> |
| 2025/07/17 | <a href="#">Rampdown Phase Two</a>                         |
| 2025/08/07 | <a href="#">Initial Release Candidate</a>                  |
| 2025/08/21 | <a href="#">Final Release Candidate</a>                    |
| 2025/09/16 | <a href="#">General Availability</a>                       |

- Stuff for newbies / prototyping
- Performance
- More concurrency goodies
- Better observability & profiling
- And more

**ONE DOES NOT SIMPLY**

**TEACH JAVA PROGRAMMING**

```
void main() {  
    IO.println("Hello, there!");  
    var name = IO.readLine("What is your name?");  
    IO.println("Nice to meet you, " + name);  
}
```

```
void main() {  
    IO.println("Hello, there!");  
    var name = IO.readLine("What is your name?");  
    IO.println("Nice to meet you, " + name);  
}
```

```
def main():  
    print("Hello, there!")  
    name = input("What is your name? ")  
    print("Nice to meet you, " + name)
```

```
void main() {  
    IO.println("Hello, there!");  
    var name = IO.readLine("What is your name?");  
    IO.println("Nice to meet you, " + name);  
}
```

```
def main():  
    print("Hello, there!")  
    name = input("What is your name? ")  
    print("Nice to meet you, " + name)
```

```
if __name__ == "__main__":  
    main()
```

# Compact source files & Instance main methods

Standard feature in Java 25

**JEP 512**

# Compact source files & Instance main methods

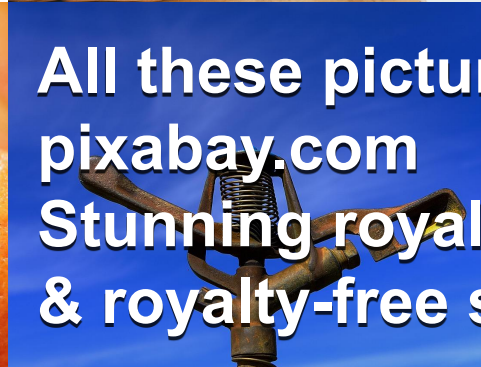
```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello World!");  
    }  
}
```

```
void main() {  
    IO.println("Hello World!");  
}
```

**DEMO TIME!**

**...WHAT COULD POSSIBLY GO  
WRONG**

makeameme.org



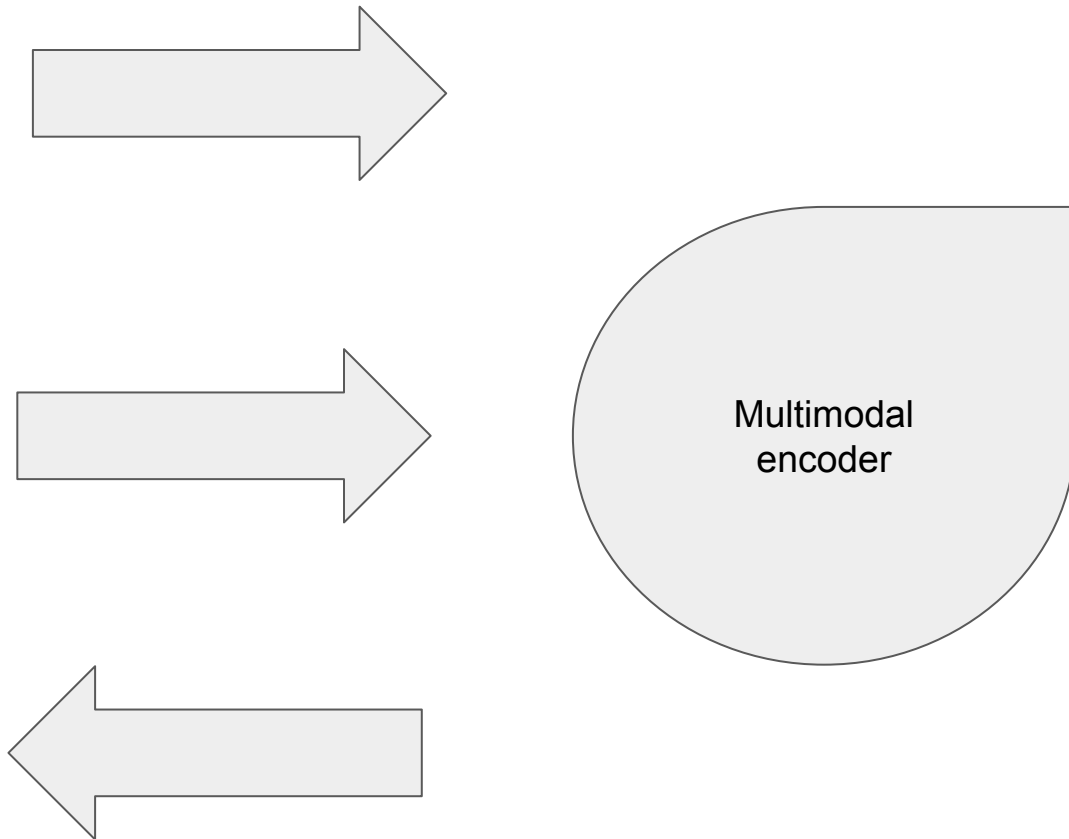
All these pictures are from  
[pixabay.com](https://pixabay.com)  
Stunning royalty-free images  
& royalty-free stock



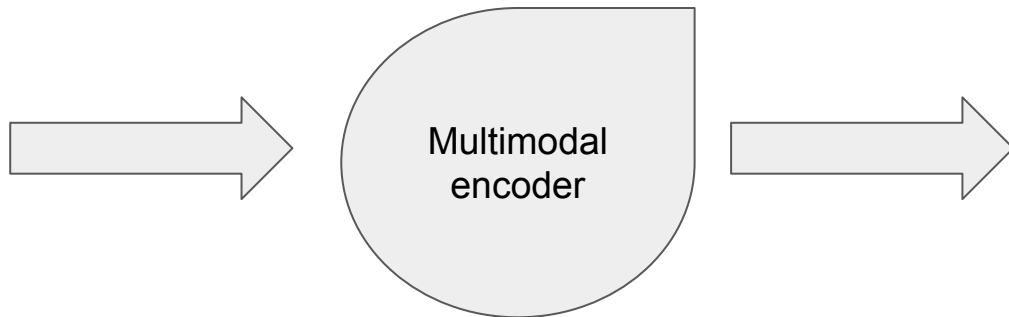


“an apple on books”

[0.12, -0.48,  
0.231, ... ]



# Preparation: Indexing

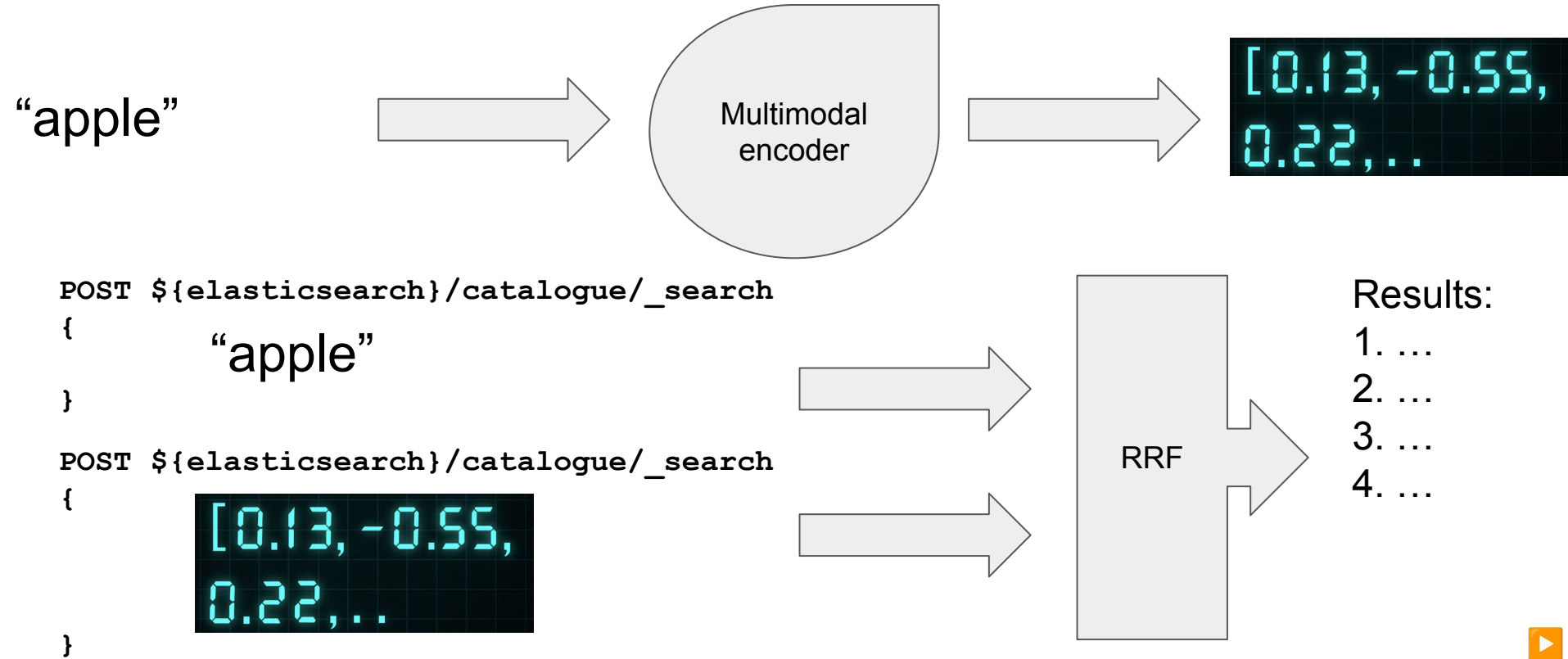


[0.12, -0.48,  
0.231, ...]

```
POST elasticsearch/catalogue
```

```
{  
  "filename" : "apple.png",  
  "embedding" : [0.12, -0.48, 0.0231, ...],  
  ...  
}
```

# Application: Searching



# Records

Standard feature in Java 16  
**JEP 395**

immutable  
transparent  
data carriers



```

final class QueryWithVector {
    private final String query;
    private final List<Float> vector;

    QueryWithVector(String query, List<Float> vector) {
        this.query = query;
        this.vector = vector;
    }

    @Override
    public String toString() {
        return "QueryWithVector[" +
            "query=" + query + ", " +
            "vector=[" + vector.getFirst()
                + ", ..., "
                + vector.getLast() + ']';
    }

    public String getQuery() {
        return query;
    }

    public List<Float> getVector() {
        return vector;
    }

    @Override
    public boolean equals(Object obj) {
        if (obj == this) return true;
        if (obj == null || obj.getClass() != this.getClass()) return false;
        var that = (QueryWithVector) obj;
        return Objects.equals(this.query, that.query) &&
            Objects.equals(this.vector, that.vector);
    }

    @Override
    public int hashCode() {
        return Objects.hash(query, vector);
    }
}

```

```

record QueryWithVector(String query, List<Float> vector) {

    @Override
    public String toString() {
        return "QueryWithVector[" +
            "query=" + query + ", " +
            "vector=[" + vector.getFirst()
                + ", ..., "
                + vector.getLast() + ']';
    }
}

```

# Module imports

Standard feature in Java 25

**JEP 511**

import all packages exported by a module with one statement



```
import java.io.IOException;
import java.math.BigDecimal;
import java.math.RoundingMode;
import java.net.URI;
import java.net.http.HttpClient;
import java.net.http.HttpRequest;
import java.net.http.HttpResponse;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.HashMap;
import java.util.List;
import java.util.Map;
import java.util.Objects;
import java.util.Scanner;
import java.util.concurrent.CompletableFuture;
import java.util.concurrent.CompletionException;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.function.Function;
import java.util.stream.Collectors;
```

```
import module java.base;
```

# Compact source files & Instance main methods

Standard feature in Java 25  
**JEP 512**

Also comes with simple IO



# Stream Gatherers

Standard feature in Java 24

**JEP 485**

# Stream Gatherers

`Stream.of(...)`

custom

`.gather(a)`

`[.gather(b)]`

intermediate

`[.gather(c)]`

operations

`.collect(...);`

Stream Gatherers - Deep Dive with the Expert: [https://youtu.be/v\\_5SKpfl2U](https://youtu.be/v_5SKpfl2U)

Gatherers: The API Your Stream Was Missing: [https://youtu.be/oVdWfU\\_IObY](https://youtu.be/oVdWfU_IObY)

Hunting with Stream Gatherers: <https://youtu.be/rvW8tu1n5x4>

Extending Functional Pipelines with Gatherers: <https://www.youtube.com/live/dxVDiJsyu3Y>



# Preview features

```
--enable-preview
```

**JEP 12**



# Structured Concurrency



Preview in Java 27

**JEP 533**

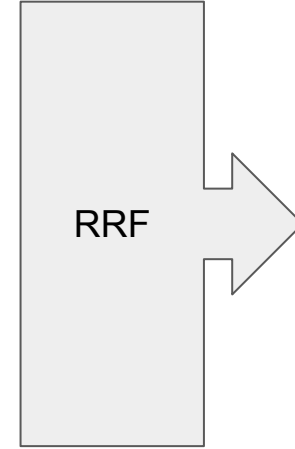
# Structured Concurrency

- better "idioms" for multi-threaded code
- synchronous way of thinking
- help eliminate thread leaks and cancellation delays

# Structured Concurrency

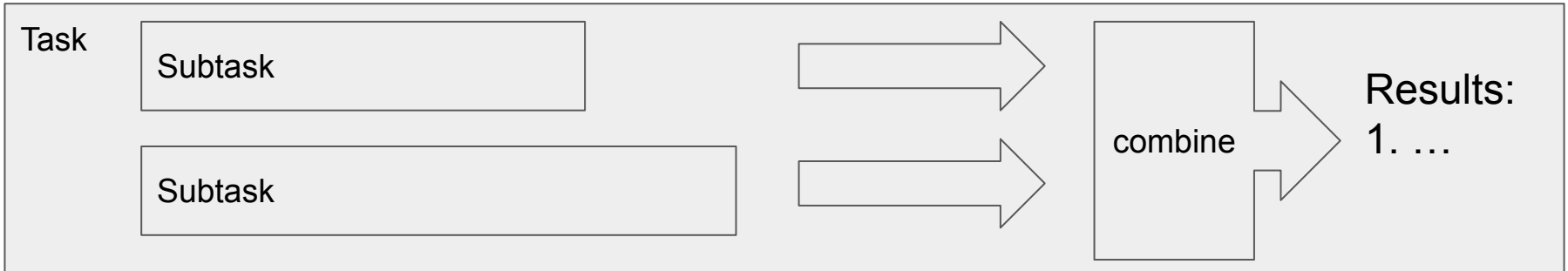
```
POST ${elasticsearch}/catalogue/_search
{
  "apple"
}
```

```
POST ${elasticsearch}/catalogue/_search
{
  [0.13, -0.55,
  0.22, ...
}
```



Results:

1. ...
2. ...
3. ...
4. ...



# Structured Concurrency

TL;DR: `.open()` with `Joiner` and `Config`,  
can (re)use thread pools, also platform ones

```
try (var scope = StructuredTaskScope.open(
  StructuredTaskScope.Joiner.awaitAllSuccessfulOrThrow(),
  config -> config
    .withName("myStructuredConcurrency")
    .withTimeout(Duration.ofSeconds(5))
    .withThreadFactory(myThreadFactory))) {
  var subtask1 = scope.fork(() → action(...));
  var subtask2 = scope.fork(this::anotherAction);
  scope.join();
  combineResults(subtask1.get(), subtask2.get());
}
```

# CompletableFuture

has no future!

**COMPLETABLE  
FUTURE**



**VIRTUAL  
THREADS**



# Lazy Constants



Preview in Java 27

**JEP 531**

- Immutable
- Thread safe
- Lazily evaluated



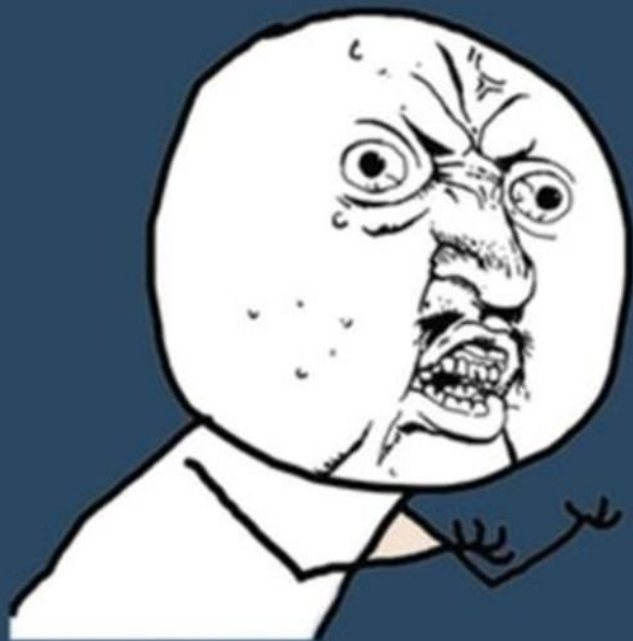
# Lazy Constants

```
static final LazyConstant<ObjectMapper> OBJECT_MAPPER =  
LazyConstant.of(ObjectMapper::new);
```

```
// and later on
```

```
OBJECT_MAPPER.get() ...  
OBJECT_MAPPER.orElse(...)  
OBJECT_MAPPER.isInitialized()
```

**Y U NO**



**RTFM?**

# Markdown Documentation Comments

Standard feature in Java 23

**JEP 467**

- Easier to read & write

Markdown in Java Docs? Shut Up and Take My Comments!

<https://blog.jetbrains.com/idea/2025/04/markdown-in-java-docs-shut-up-and-take-my-comments/>



```
/// Please remember to run with a VM option
/// `--add-modules jdk.incubator.vector`
// For more details, please see [JEP-489](https://openjdk.org/jeps/489).
```

```
public class Operations {
```

```
    public static
```

```
        // Use a
```

```
        int size
```

```
        double[]
```

```
        double[]
```

```
        double[]
```

```
        // Initi
```

```
        for (int i = 0; i < size; i++)
```

org.przybyl

public class Operations

Please remember to run with a VM option

--add-modules jdk.incubator.vector

For more details, please see [JEP-489](https://openjdk.org/jeps/489).

computations



rmance differences.

# Stay up to date

1. pom.xml / gradle.kts
2. new idioms



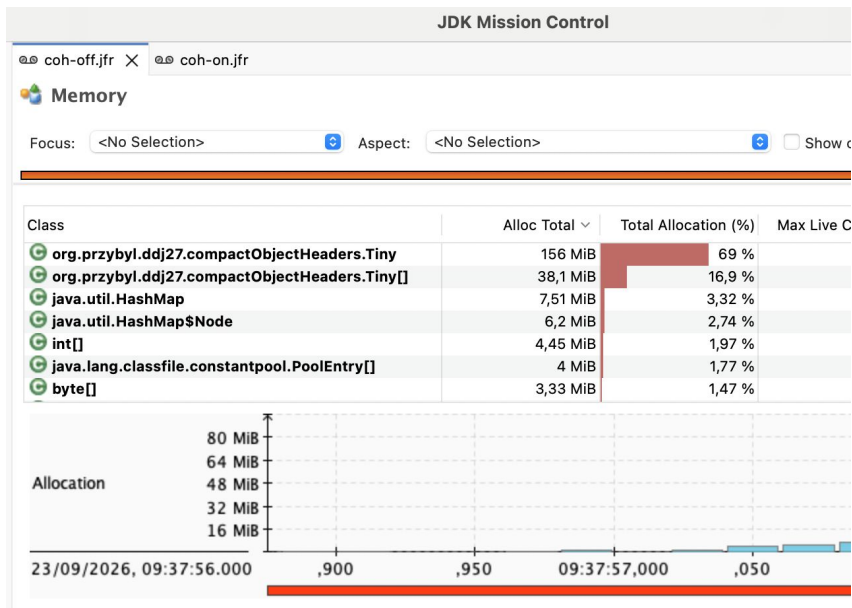
# Compact Object Headers by Default

Standard feature in Java 27

**JEP 534**

- No need to do anything in Java 27

```
final class Tiny {}
int count = 10_000_000;
var objects = new Tiny[count];
for (int i = 0; i < count; i++) {
    objects[i] = new Tiny();
}
```



- Example: [youtrack.jetbrains.com/issue/IJPL-212987](https://youtrack.jetbrains.com/issue/IJPL-212987)

# Pattern matching with primitive types



Preview in Java 27

**JEP 532**

- Allow primitive types in all pattern contexts
- Extend `instanceof` and `switch` to work with all primitive types



```
int memberDiscountPercentage(boolean isGoldMember) {  
    return isGoldMember ? 20 : 5;  
}
```

```
int itemDiscountPercentage(int items) {  
    if (items == 2) {  
        return 5;  
    } else if (items == 3 || items == 4) {  
        return 10;  
    } else if (items ≥ 5) {  
        return 20;  
    }  
    return 0;  
}
```

```
int memberDiscountPercentage(boolean isGoldMember) {  
    return switch (isGoldMember) {  
        case true → 20;  
        case false → 5;  
    };  
}
```

```
int itemDiscountPercentage(int items) {  
    return switch (items) {  
        case 2 → 5;  
        case 3, 4 → 10;  
        case int i when i ≥ 5 → 20;  
        default → 0;  
    };  
}
```

# PROGRAMMERS: Know the difference!

```
((() => {  
  function getName(gender: String = ""): String {  
    if(gender == "male") {  
      return "Robert";  
    }  
    if(gender == "female") {  
      return "Margot";  
    }  
    return "";  
  }  
  
  const gender = "male";  
  const name = getName(gender);  
  
  console.log(`The child's name is ${name}`);  
})();
```

If statement



Switch case



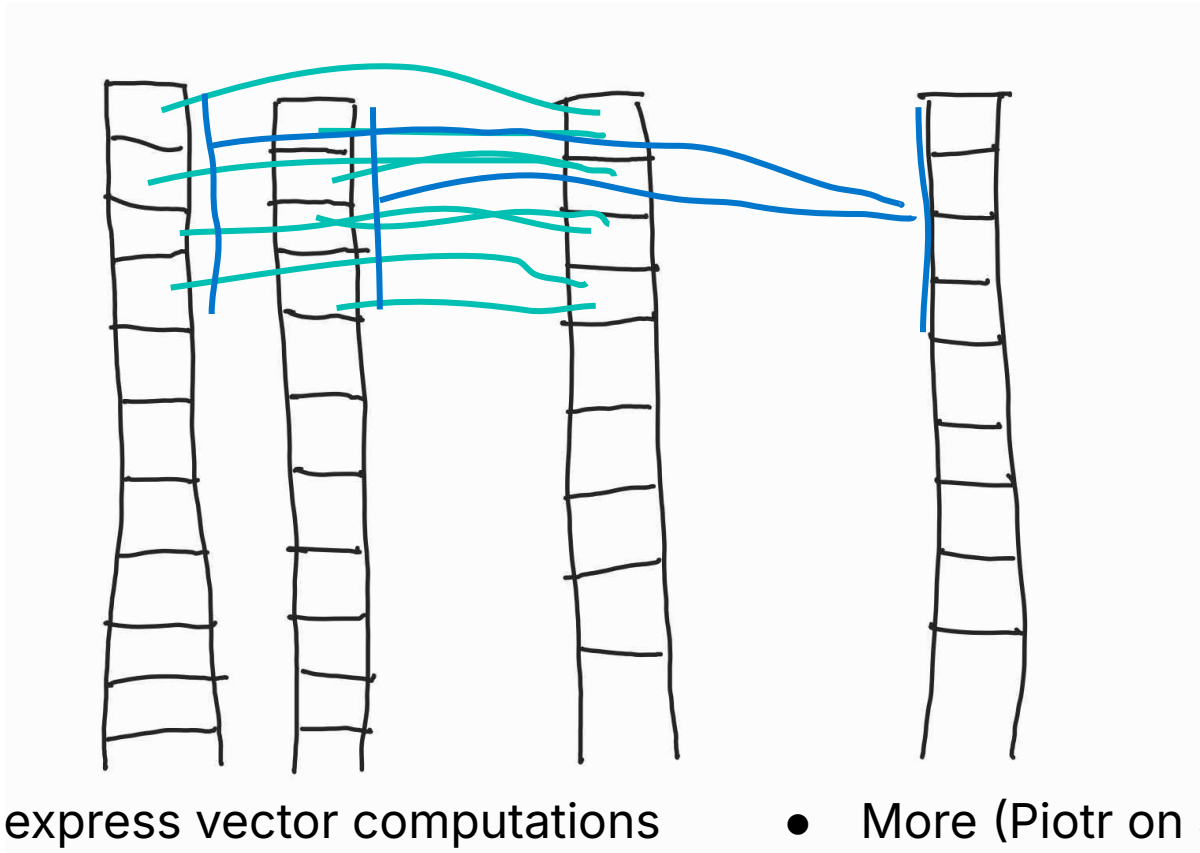
# Vector API (Panama)



Incubator in Java 27

**JEP 537**





- API to express vector computations
- Different performance characteristics
- More (Piotr on SIMD in Java):  
[vimeo.com/1223454521](https://vimeo.com/1223454521)

**CAN'T DO BADLY AT A BENCHMARK**

**IF YOU CREATE YOUR OWN**

```
$ docker exec -it es-local-dev /bin/sh -c "ps aux | cat"
/usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60
-Des.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss1m -Djava.awt.headless=true
-Dfile.encoding=UTF-8 -Djna.nosys=true -XX:-OmitStackTraceInFastThrow -Dio.netty.noUnsafe=true
-Dio.netty.noKeySetOptimization=true -Dio.netty.recycler.maxCapacityPerThread=0
--add-opens=org.apache.lucene.core/org.apache.lucene.codecs.lucene99=org.elasticsearch.server
--add-opens=org.apache.lucene.core/org.apache.lucene.codecs.hnsw=org.elasticsearch.server
--add-opens=org.apache.lucene.core/org.apache.lucene.internal.vectorization=org.elasticsearch.server
-Dlog4j.shutdownHookEnabled=false -Dlog4j2.disable.jmx=true -Dlog4j2.formatMsgNoLookups=true
-Djava.locale.providers=CLDR -Dorg.apache.lucene.vectorization.upperJavaFeatureVersion=26
-Des.path.home=/usr/share/elasticsearch -Des.distribution.type=docker -Des.java.type=bundled JDK
--enable-native-access=org.elasticsearch.nativeaccess,org.apache.lucene.core
--enable-native-access=ALL-UNNAMED --illegal-native-access=deny -Des.cgroups.hierarchy.override=/
-XX:ReplayDataFile=logs/replay_pid%p.log -XX:+EnableDynamicAgentLoading
-Djdk.attach.allowAttachSelf=true
--patch-module=java.base=/usr/share/elasticsearch/lib/entitlement-bridge/elasticsearch-entitlement-bridge-9.4.0.jar
--add-exports=java.base/org.elasticsearch.entitlement.bridge=org.elasticsearch.entitlement,java.logging,java.net.http,java.naming,jdk.net,jdk.zipfs -XX:+UseG1GC
-Djava.io.tmpdir=/tmp/elasticsearch-1825357143797079744 --add-modules=jdk.incubator.vector
-Dorg.apache.lucene.store.defaultReadAdvice=normal
-Dorg.apache.lucene.store.MMapDirectory.sharedArenaMaxPermits=1 -XX:+HeapDumpOnOutOfMemoryError
-XX:+ExitOnOutOfMemoryError -XX:ErrorFile=hs_err_pid%p.log
-Xlog:gc*,gc+age=trace,safepoint:file=gc.log:utctime,level,pid,tags:filecount=32,filesize=64m -Xms128m
-Xmx2g -XX:MaxDirectMemorySize=1073741824 -XX:G1HeapRegionSize=4m
-XX:InitiatingHeapOccupancyPercent=30 -XX:G1ReservePercent=15 --module-path
/usr/share/elasticsearch/lib --add-modules=jdk.net --add-modules=jdk.management.agent
--add-modules=ALL-MODULE-PATH -m org.elasticsearch.server/org.elasticsearch.bootstrap.Elasticsearch
```



# What did we play with?

Vectors from Panama

Records

Structured Concurrency

Stream Gatherers

Lazy Constants

Compact Object Headers

Pattern Matching

Compact Classes, module imports, simple IO

... and more

Each new Java version is:

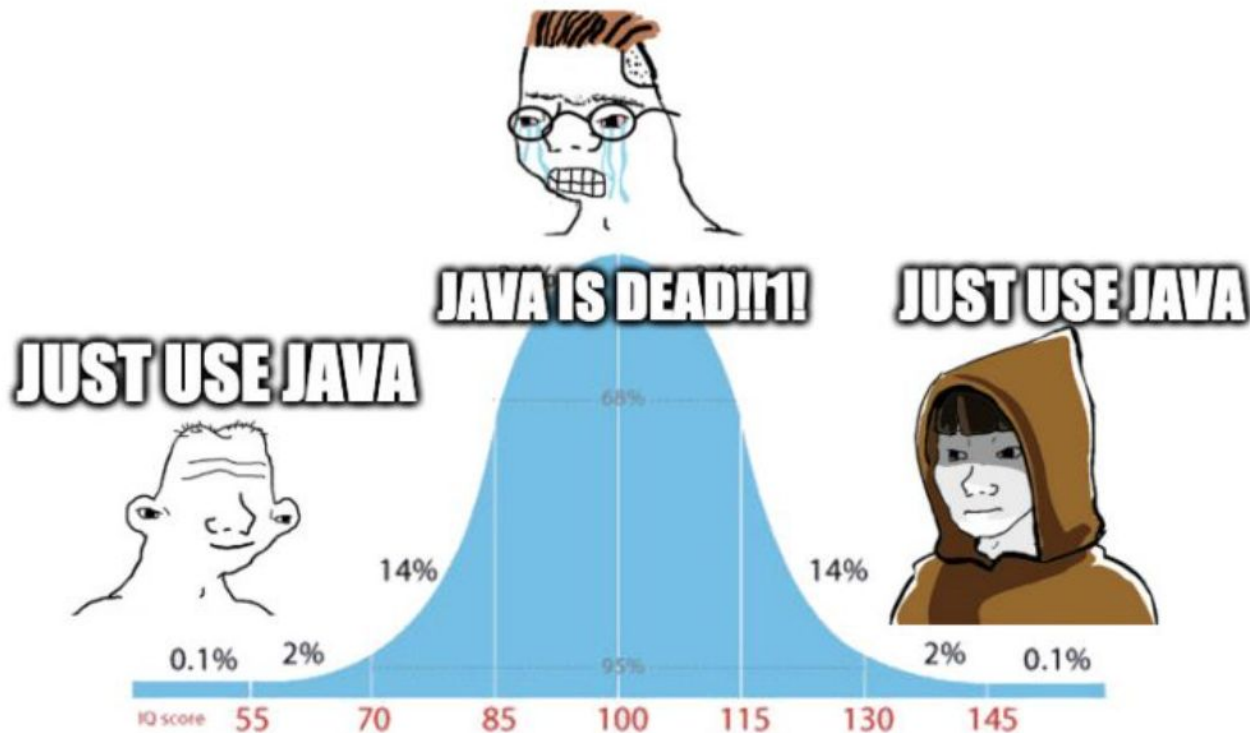
- easier
- faster
- better
- and (guess what?) still free

Why not use it?



**Gunnar Morling** • 1st  
Technologist at Confluent  
6d • 🌐

Java is dead, just sayin'...



Feedback, please!



GOTO session



How was it?



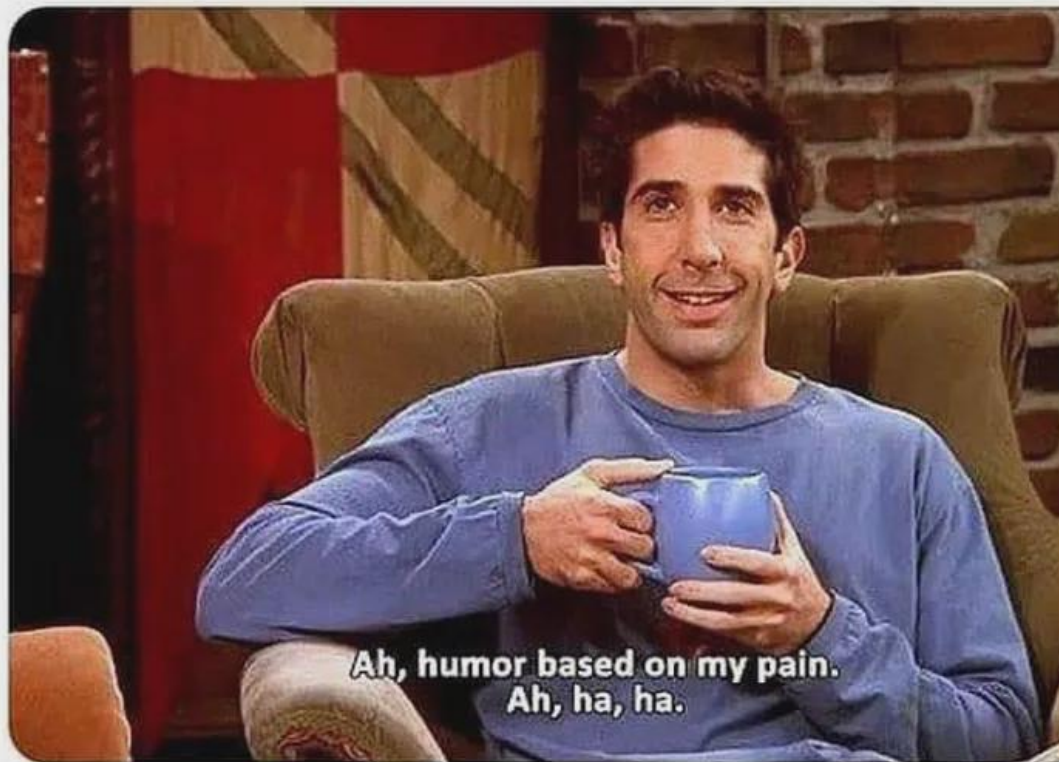
[bit.ly/LMJPW-GOTO](https://bit.ly/LMJPW-GOTO)

The (QR) code & everything



[bit.ly/LMJPW-all](https://bit.ly/LMJPW-all)

# Programmers looking at programming memes





Piotr Przybył

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@piotrprz.bsky.social

<https://softwaregarden.dev>

Tusind tak

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<https://maritvandijk.com>

This meme has been left  
as an exercise for the  
reader.